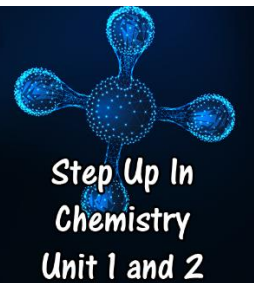


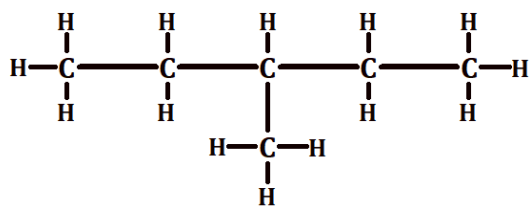
3.2 Hydrocarbons Structure and Naming

Problems Worksheet



1. Hydrocarbons contain only carbon and hydrogen atoms.
 - a. What about carbon gives it the ability to form an essentially unlimited number of different hydrocarbons?
 - b. What is the different between alkanes, alkenes and alkynes?
 - c. What is IUPAC nomenclature?
2. Describe the following properties of hydrocarbons:
 - a. Boiling points
 - b. Solubility

3. Draw the following compound in each of these different ways:



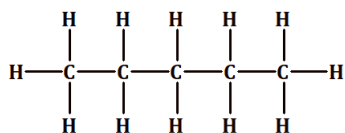
a. Molecular formula

b. Condensed structural formula

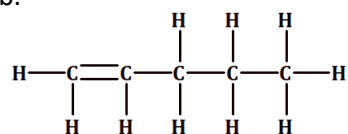
c. Line-angle formula

4. Name the following hydrocarbons:

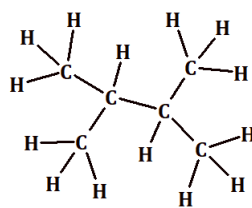
a.



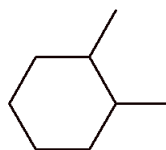
b.



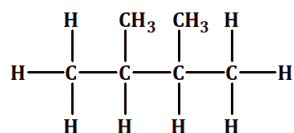
i.



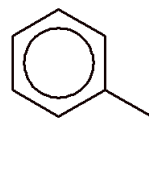
j.



c.



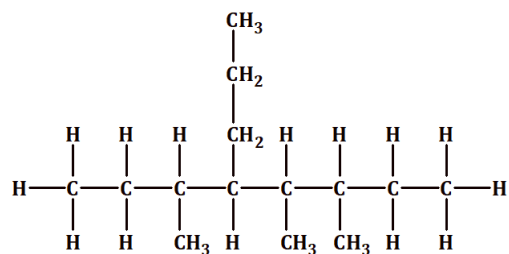
k.



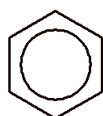
d.



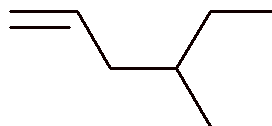
l.



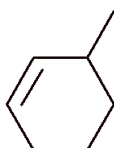
e.



m.



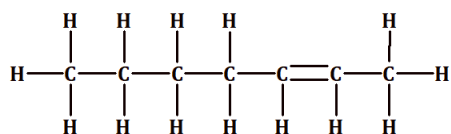
f.



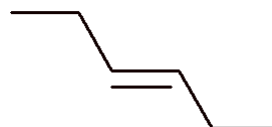
n.



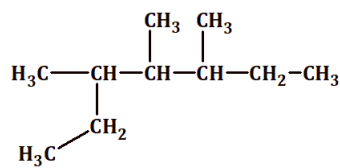
g.



o.



h.



5. Draw structural formulas for the following:

a. 2-methylbutane

b. 2,4-methylpentane

c. 2-ethylpent-1-ene

d. cis hept-3-ene

e. trans pent-2-ene

f. 1,2-dimethylcyclopentane

g. propylbenzene

h. 4,5dimethylclohexene

6. Most hydrocarbons have isomers. Describe the following types of isomers:
- a. Chain isomers
 - b. Position isomers
 - c. Geometric isomers
7. Use line-angle formulas to show four different isomers with the following molecular formulae:
- a. C_6H_{14}
 - b. C_4H_8